https://doi.org/10.1038/s41551-025-01363-2

Coordinated AI agents for advancing healthcare

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Decentralized yet coordinated networks of specialized artificial intelligence agents, multi-agent systems for healthcare (MASH), that excel in performing tasks in an assistive or autonomous manner within specific clinical and operational domains are likely to become the next paradigm in medical artificial intelligence.

Specialist artificial intelligence (AI) models are being developed or deployed for most tasks in healthcare, from clinical applications such as risk prediction and patient monitoring to non-clinical tasks such as streamlining hospital operations, scheduling appointments and processing claims. The trend is towards foundation models with generalist capabilities that dynamically adapt to novel tasks and that accommodate flexible multimodal inputs. However, nascent generalist models remain specialist models in the wider biomedical domain: they operate independently from one another and do not account for broader contexts. These shortcomings could limit the capabilities of generalist models and result in conflicting recommendations.

The integration of multiple state-of-the-art large language models (LLMs) helps to transform these models into powerful AI agents with several unique capabilities: specifically, natural language communication, planning and reasoning over vast amounts of unstructured data, the use of internet and database searches, the navigation of graphical user interfaces, and the creation of code. Indeed, LLM-based agentic and multi-agentic systems have outperformed standard and more sophisticated LLM-prompting techniques (such as chain of thought) in several complex tasks. Such performance improvements appear to emerge with more advanced LLMs and scale with the number of agents[1].

To increase model performance, developers of LLMs have primarily relied on larger numbers of parameters in the models, on bigger training datasets and on more train-time compute. Some of the latest model architectures (such as that of the o1 model from OpenAI) also scale at the time of inference (test-time compute scaling), owing to a built-in chain-of-thought process for 'thinking and reasoning' before responding in a manner akin to slow yet logical thinking (rather than fast and instinctual thinking), yet at the expense of increased token use and latency[2,3]. With other models, it has been shown on an abstract-reasoning benchmark that model weights can be adjusted temporarily during inference (test-time training) to advance performance[4]. The expanded capabilities of these newer architectures will increase the performance of multi-agentic models and will be crucial to achieving the completion of complex healthcare tasks with high reliability.

AI agents function within the autonomous–assistive spectrum of tasks. Autonomous AI agents can work independently to collect and process data and take actions within their designated areas of expertise. For example, an autonomous AI agent in radiology might analyse medical images to detect signs of disease and automatically generate a report. Assistive AI agents support human decision-making by providing insights, recommendations or suggestions to healthcare providers, who then make the final decisions on the basis of their expertise and the AI-generated information. For example, a conversational agent powered by an LLM or a vision–language model trained on a massive corpus of biomedical literature and current guidelines can provide expert-level answers to medical questions.

The seamless coordination between decentralized multi-agent systems for healthcare (MASH) is a formidable challenge[5,6]. The agents must be capable of communicating effectively to share insights and coordinate actions. The output of the MASH network ought to be reliable and must avoid propagating errors or biases from individual agents through the network. The roles and responsibilities of autonomous and assistive agents must be delegated carefully, depending on task complexity[6], on risk level and on regulatory and ethical considerations surrounding AI in medical decision-making. Several multi-agentic frameworks in domains other than healthcare are addressing these challenges. Notably, one healthcare-specific multi-agentic architecture consisting of a primary agent and several sub-specialized support agents (including specialists for medications, labs and vitals, hospital and payor policy, electronic health records, and privacy and compliance) has outperformed humans on several measures in multi-turn voice conversations[7].

In this Comment, we argue that a MASH network could function as a distributed form of artificial general intelligence in healthcare that addresses each patient's unique needs to deliver high-quality care. In the following sections, we outline what this multi-AI-agent-powered future might hold through the journey of a patient navigating a healthcare system (Figs. 1–4). As Supplementary information, we provide additional references to relevant literature.

Seamless coordination

In the science-fiction movie 'Her' released in 2013, AI assistants develop a novel language to communicate more efficiently among themselves. This fictional portrayal highlights the potential for AI agents to create optimized communication methods that may be unintelligible to humans. However, in healthcare, we anticipate that the trend will be in the opposite direction: human-understandable interactions between AI agents, to ensure interpretability, accountability and seamless integration with human workflows.

Traditionally, the interfaces between different AI agents have been engineered using standardized protocols and application programming interfaces. These technical solutions enable basic communication between agents and have been essential for the exchange

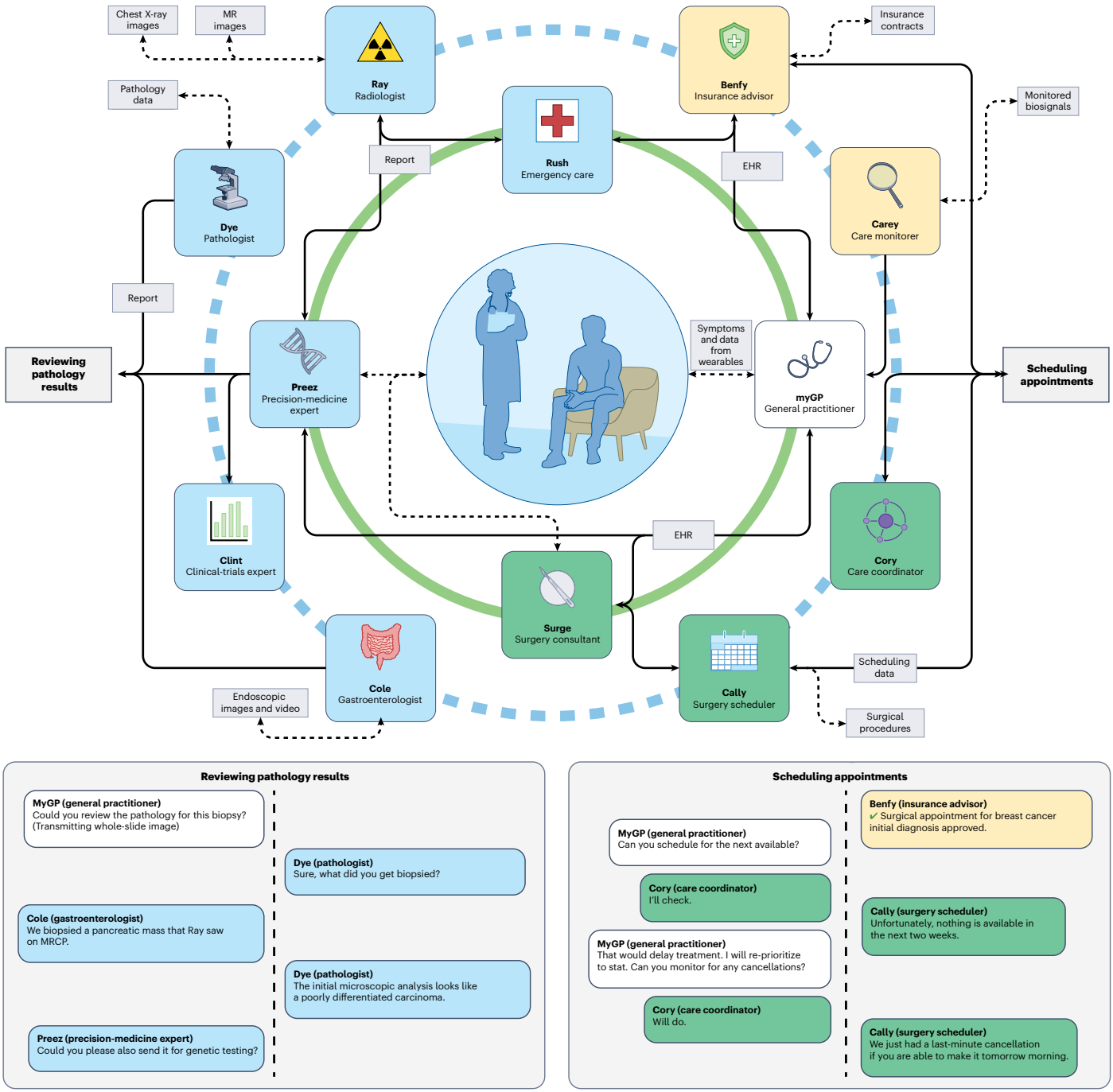

Fig. 1 | Specialized AI agents in a MASH network coordinate to accomplish tasks within the broader healthcare environment. The patient–provider interaction is supported by agents with frequent direct human interaction (green circle) and those with less frequent human interaction (blue dashed circle). The patient interacts with the MASH network through their personal general practitioner. Data are transferred between agents (black solid lines) or between agents and physical-world sources such as the patient, provider or diagnostic and therapeutic equipment (black dashed lines). Inter-agent communication occurs primarily through natural language, and only the information necessary to carry out the particular task is accessed and shared. EHR, electronic health record; MRCP, magnetic resonance cholangiopancreatography.

of structured data. However, as AI agents become increasingly sophisticated, more flexible and intuitive, interfaces that can also handle the complexity of healthcare data and decision-making processes will be needed. This could be achieved by harnessing the powerful natural-language-processing capabilities of LLMs to enable AI agents to communicate using human language. By allowing agents to interact

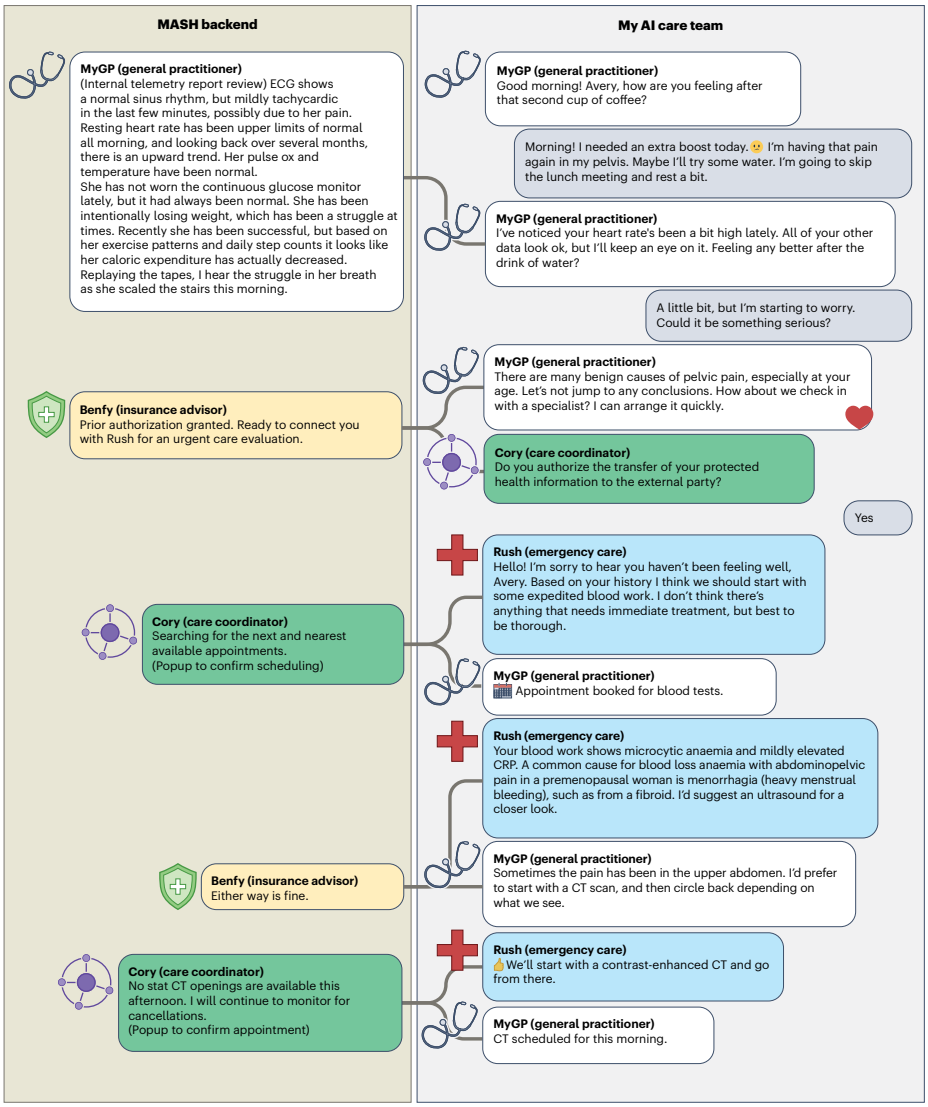

Fig. 2 | An envisioned chat among agents in a MASH network during initial presentation and evaluation. The agents are engaged in the evaluation and management of an acute-care episode, including a consultation with an urgent-care specialist, the scheduling of laboratory and imaging tests, and the interpretation of the results with follow-up coordination. Similarly to current group-chat interfaces, users (patients and clinicians) would interact with the MASH system through text or speech via a smartphone or wearable device. The backend agent-to-agent log of the chat would be available for performance monitoring and validation. ECG, electrocardiogram; CRP, C-reactive protein; CT, computed tomography.

with each other and with humans using the same language, MASH can become more accessible. Indeed, the agents would be more easily integrated into existing healthcare workflows, with healthcare professionals interacting as intuitively and seamlessly as they would with a human colleague.

Natural language interfaces would also allow for more flexible and adaptable communication between AI agents. As new agents are developed or existing ones are updated, they can be integrated into a MASH network without the need for extensive reengineering of application programming interfaces or protocols. Such scalability is essential in medical AI and particularly for the proliferation of commercial products.

Moreover, by using human-like language to convey insights and recommendations, AI agents would increase the interpretability and accountability of their decisions. Inter-agent natural-language communication can be easily logged and audited, creating a clear record of the decision-making process. Should an adverse event occur, the full history of agent interactions can be reviewed by humans. Such enhanced provenance is crucial for building trust in MASH among healthcare providers and patients, who would then better understand the reasoning behind the suggestions of the MASH in the contexts of success and failure.

Decentralized networks

One key consideration in a network of AI agents in healthcare is that each agent be trained and operated on data sources different from those of other agents. Because privacy of protected health information is a chief concern with digital health tools, a centralized AI model

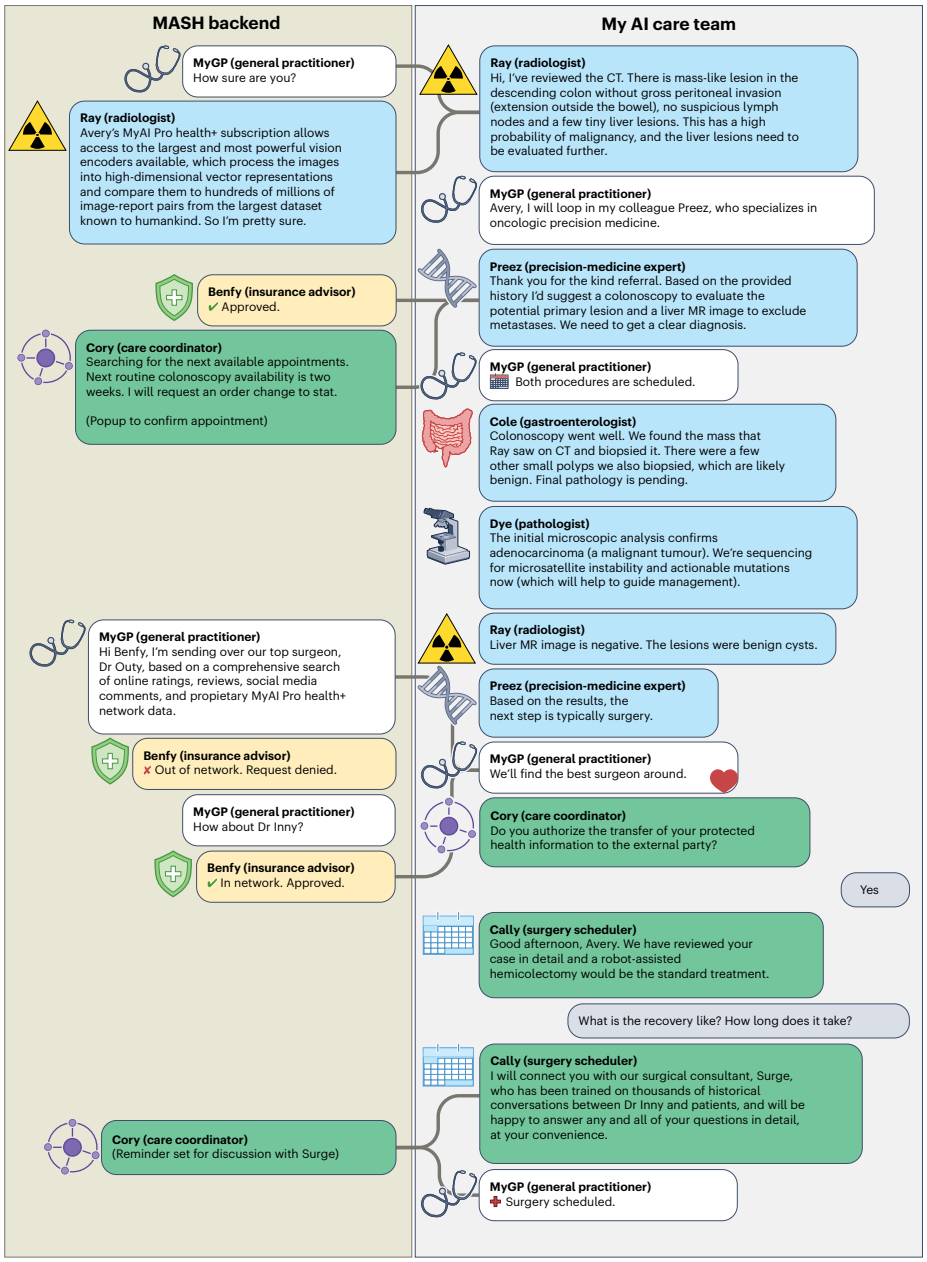

Fig. 3 | Continued envisioned chat among agents in a MASH network during diagnosis and treatment planning. The agents are engaged in the evaluation and management of the acute-care episode, including consultation with multiple specialty agents and a human physician, and the navigation of insurance coverage.

trained on all patient data would pose potentially catastrophic risks to patient privacy. In addition, AI systems trained on similar data and in similar ways (known as 'algorithmic monoculture') may be prone to similar biases and fail in similar ways. A decentralized network of agents trained and operating on different data sources could thus allow healthcare organizations to leverage multiple specialized AI models without extensive data sharing or centralized control while providing helpful diversity and oversight.

In a decentralized approach, each AI agent can specialize in a particular healthcare domain or function, such as diagnostics, treatment planning or patient monitoring. The agents can be trained on specific datasets relevant to their area of focus, ensuring access to the information needed to perform their tasks effectively, without unnecessarily accessing or storing the entire patient record. For example, a diagnostic AI agent responsible for diagnostics on the basis of medical imaging could be trained solely on a dataset of X-rays, computed tomography scans and magnetic resonance images; patient genetic information or treatment history, for instance, would not be accessible to the agent. Similarly, an AI agent focused on medication management could operate using limited data from electronic health records and pharmacy benefit management systems without requiring biometric data or lifestyle information. Any additional information needed at inference can

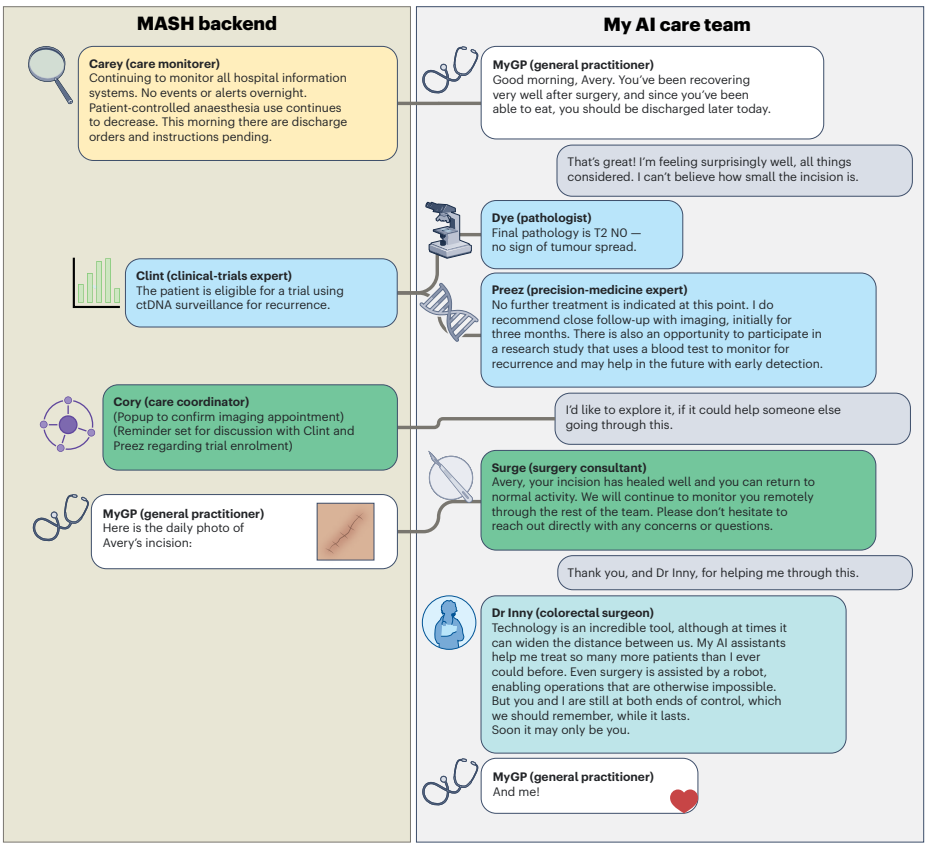

Fig. 4 | Continued envisioned chat among agents in a MASH network during treatment and follow-up. The agents are engaged during the patient's surgical admission and follow-up by monitoring the hospital's information systems, via the remote monitoring of the patient after their discharge, and through intermittent human–physician interactions. ctDNA, circulating tumour DNA.

be requested via access controls between authenticated and trusted agents in the MASH network.

Federated learning — a decentralized training set-up for AI models where each model is trained on datasets that stay within individual institutions — enables healthcare organizations to collaborate in the development of AI models with improved generalizability. However, in practice, federated learning can be challenging to implement, particularly for models with large numbers of parameters and a high communication overhead for the transfer of model weights[8]. For MASH, federated learning would allow for more efficient data processing as well as more robust generalizability and reduced privacy risks. Each AI agent would focus on analysing and interpreting the specific data relevant to its task, limiting the required model size and therefore communication overhead, as well as training and inference costs. This will require understanding of the optimal cross-domain information requirements.

It is essential to recognize that multi-agent systems come with unique risks. For example, with AI agents interacting in series (where the output of one agent serves as the input of another), errors associated with any one agent in the chain might be compounded, and the performance and reliability of the entire system would be limited by its weakest link. Inaccurate or biased results from one agent could propagate through the system in the form of a cascade of errors with potentially harmful outcomes.

Yet, a well-designed MASH network could prove more reliable and resilient than a monolithic system. Indeed, a diversity of agents would avoid 'algorithmic monoculture'. Still, there should be enough cross-domain expertise for a subset of agents to function adversarially and limit the risk of cascading errors. These sort of checks and balances have been shown in several domains. In particular, for general purpose LLMs, 'debate' between agents can improve factual accuracy and reasoning[9]. For healthcare tasks, general purpose LLMs are able to self-correct some confabulations[10], and some healthcare-specific architectures use task-specific support agents to double-check the primary model[7,11]. Dedicated quality-assurance agents could thus be designed and incorporated into a MASH architecture at various levels. Moreover, human-in-the-loop oversight (participatory) or human-on-the-loop oversight (supervisory) would be paramount until safe autonomous performance can be demonstrated.

Synergistic integration

Within MASH, each AI agent may be broadly categorized into two main types: clinical and operational. Clinical agents would focus on tasks directly related to patient care, such as diagnostics, treatment planning and disease monitoring. Operational agents would instead handle administrative and logistical aspects of healthcare delivery, such as appointment scheduling, resource allocation and billing. Although these two types of agent may rely on different data

sources and have distinct objectives, they would ultimately need to collaborate to ensure the smooth functioning of the healthcare system. Clinical agents would require input from operational agents to ensure that patients receive timely and appropriate care, and operational agents would depend on insights from clinical agents to optimize resource utilization and improve overall efficiency. For example, a clinical agent responsible for monitoring patients with chronic conditions may identify a patient who requires an urgent follow-up appointment. This information would then be passed on to an operational agent in charge of scheduling, which would prioritize the appointment according to the assessment of the clinical agent. Similarly, an operational agent managing inventory and supplies could alert clinical agents about potential shortages of critical medications or equipment so that they can adjust treatment plans accordingly.

Integrating clinical and operational agents can create powerful predictive models that can anticipate patient needs and optimize resource allocation. For instance, by combining data from electronic health records, appointment schedules and patient-flow patterns, AI agents could predict periods of high demand and proactively adjust staffing levels and resource distribution to minimize wait times and improve patient satisfaction. In the future, enhanced multimodal capabilities in the models and clearer regulatory frameworks may blur the distinction between clinical and operational agents.

Eventually, MASH may lead to considerable cost reductions by streamlining clinical and non-clinical operations. Initial implementation and scaling, however, will require substantial investment in information-technology infrastructure, such as network upgrades, cloud services and high-performance computing hardware and data-storage solutions. Additional investment will be needed for enhanced cybersecurity measures, for the training of healthcare providers and operational staff, and for the integration of MASH with existing systems. Managing integration and change will face unique challenges for the many healthcare organizations that rely on legacy systems or on outdated and idiosyncratic hardware, software or data formats, which may not be compatible with modern interoperability standards such as HL7 FHIR (health level-7 standards for fast healthcare interoperability resources). The natural language-communication abilities of LLMs will help bridge this gap, but infrastructure or technical debt will ultimately need to be paid through upgrades or further customization to ensure that MASH agents and non-AI systems work together seamlessly.

Healthcare systems will have to weigh implementation costs against improvements in long-term operational efficiency and clinical outcomes. A phased deployment, starting with pilot programmes, will be needed to validate the performance of MASH in shadow environments or other controlled environments, to gather feedback and to make necessary adjustments before broader adoption. A phased approach would also allow institutions to better understand the return on investment and to adjust strategies accordingly.

Collaborative workflows and task delegation

As AI continues to advance, physician-independent workflows are likely to emerge in certain areas of healthcare. These workflows may be driven primarily by collaborations between clinical and operational AI agents and may streamline processes, optimize resource utilization and improve patient outcomes. For example, AI-driven triage systems could assess patient symptoms, determine the urgency of care needed and direct patients to the most appropriate healthcare setting with available resources, all without the need for direct physician involvement.

However, physician-independent workflows will not be suitable for many aspects of healthcare. Complex cases, rare diseases and situations requiring nuanced clinical judgement will continue to require the expertise and experience of human physicians. In these instances, AI agents will serve as support tools for physicians. The agents may provide data-driven insights and recommendations and assist with workflows, such as the generation of patient histories from multi-turn questioning[12].

Physicians will have a critical role in overseeing and validating all clinical AI workflows to ensure their continuous safety and alignment with evidence-based practices. This will require physicians to develop a deep understanding of the capabilities and limitations of AI systems. Healthcare and professional organizations must invest in education and training initiatives to prepare physicians for the AI-augmented landscape. This includes developing skills in data analysis, AI-system management and collaborative decision-making. These topics should be integrated into medical licensing and specialty-board examinations, into continued medical education and even into hospital credentialling.

Moreover, there must be a culture of collaboration and trust between physicians and AI systems. The digital workflows in the electronic health record have been a major contributor to physician dissatisfaction and burnout[13]. Understanding the optimal delegation of tasks and the design of human–AI user interfaces to maximize effectiveness while minimizing physician burnout will be critical. The alignment of these systems to serve patients and healthcare professionals would require the direct involvement of physicians in the development, implementation and evaluation of AI-driven workflows.

Regulation and benchmarks

Regulatory processes for AI-driven healthcare technologies should be rigorous and evidence-based to ensure patient safety and system efficacy. Regulatory bodies such as the United States Food and Drug Administration, the UK Medicines and Healthcare products Regulatory Agency and the European Medicines Agency are developing frameworks for the evaluation and approval of AI-based medical devices. Specific considerations for devices and tools leveraging LLMs have been suggested[14]. However, these frameworks are primarily designed for standalone software as a medical device and may not directly apply to multi-agent systems.

A possible regulatory framework for AI agents would be to approach them similarly to how it is done with human providers: by granting progressively increasing responsibility and autonomy as their training and experience deepens. This could include continuous performance evaluation by the certifying body as well as recertification at various time points. This would be analogous to the progression of a physician through medical school, internship, residency, fellowship and continuing medical education.

However, the multiagent component of MASH presents unique challenges. In particular, traditional evaluation metrics for machine-learning models and single-agent AI systems do not capture the complexities of inter-agent interactions and decentralized coordination. New metrics tailored to MASH must be developed to assess individual agent performance as well as the agents' collective behaviour, including their coordination efficiency, resilience to cascading errors, data security, absence of bias and combined impact on patient outcomes. Both validation of each individual agent and system-level validation will be necessary.

Several general purpose multiagent benchmarks have been proposed[5,15], including a healthcare-specific multiagent benchmark using simulated clinical encounters that included the intentional introduction of biases into the agents[16]. Future MASH benchmarks should include complex and diverse tasks that bridge clinical and operational domains, to ensure efficacy, as well as 'red teaming', to ensure privacy and safety. Developing these benchmarks will require close collaboration among AI researchers, healthcare professionals and regulatory bodies. The safe and effective integration of MASH systems into healthcare environments will require the proactive establishment of clear standards to objectively evaluate the systems individually and collectively.

The unexpected capabilities of LLM agents

The conventional value proposition of AI in healthcare has centred on reducing administrative burdens to allow medical professionals to focus on patient interactions and on augmenting human intelligence to provide more accurate and precise care. However, advancements in LLM-based agents have also demonstrated the surprising ability of AI to elicit empathy in humans[17]. Indeed, in diverse healthcare tasks, interactions with text-only and text-to-speech LLMs were rated as more empathetic than human-to-human interactions[18]. In text-to-speech tasks, the degree of reported empathy appears to be inversely proportional to the response latency, which is a technical challenge. In other domains, affective visual avatars have been developed for agents. Rapid technological advancement is likely to improve the empathic capabilities of such AI systems.

An agentic approach could also help solve the human–AI alignment problem. Analogous to the rapport developed between patients and their healthcare providers, a personal AI agent with longitudinal engagement will gain invaluable feedback and historical knowledge from the patient. Because a deep understanding of the patient's values and beliefs, prior experiences and communication preferences via reinforcement learning will be essential when the patient interfaces with the broader MASH network, and particularly for interactions among autonomous agents that perform tasks without a human in the loop, a personalized agent would mirror the human within the system and function as the human's trusted intermediary to the system. In other words, trusting a personal agent would imply trusting the MASH network. And trusting a personal AI agent that is patient and attentive may become easy.

The future of synergetic AI agents in healthcare

In Sir Luke Fildes' iconic painting 'The Doctor' (1891), a physician sits pensively by the bedside of a sick child, epitomizing the dedication, empathy and attentiveness that have long been the hallmarks of compassionate healthcare. In the new era of medical AI, we can envision a future where this scene is transformed by the presence of an ever-present network of specialized AI agents working in harmony to support the patient and the doctor.

Multi-agent systems for healthcare are likely to be the next paradigm in medical AI. We expect that it is likely that within the next decade there will be AI agents that monitor sensor data, analyse medical histories, provide real-time insights and regularly collaborate with each other to provide care and to monitor procedures and outcomes. In some cases, the agents will communicate directly with the physician or the patient to offer guidance, answer questions or provide emotional support. Natural-language interfaces will enable these interactions to be as seamless and intuitive as conversing with a knowledgeable expert. In other instances, the agents will operate behind the scenes, silently coordinating resources and personalized treatment plans, all in service of providing the best possible care.

We expect that the collective intelligence of MASH will ultimately unlock a distributed form of healthcare that democratizes personalized, precise and proactive patient care. We hope that workflows are streamlined, that unseen patterns in vast amounts of data become visible and that physicians will be augmented with critical information. Although the scene in 'The Doctor' will serve as a powerful reminder that human emotion remains at the heart of medicine, the attentive gaze of the physician will be mirrored by the watchful eyes of MASH, intently focused on both the patient and the doctor, working tirelessly to help create a seamless healthcare experience where every interaction is imbued with empathy and compassion while improving outcomes. Such a shift in medical paradigm will redefine what it means to provide healthcare in the age of coordinated AI.

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Published online: 1 April 2025

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Acknowledgements

We thank R. Reddy for early review and discussion of the figures.

Author contributions

M.M., E.T. and P.R. all contributed to the design, drafting and editing of the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41551-025-01363-2.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Peer review information *Nature Biomedical Engineering* thanks Ahmed Alaa and Vivek Natarajan for their contribution to the peer review of this work.